

COMPSCI 752

Big Data Management

Strategic Exercise 2

XPath

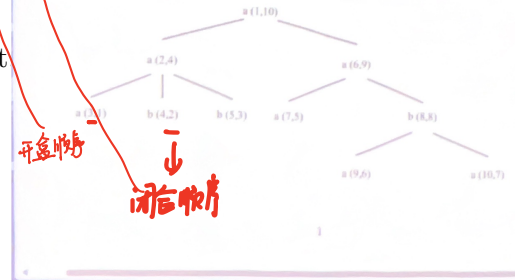
Application domain. For questions 1, 2, and 3, consider the following XML document:

```
<?xml version="1.0" encoding="UTF-8"?>
<a><a><a/><b/><b/></a><a><a/><b><a/><a/></b></a></a>
```

Handwritten notes: Green box around the XML declaration and the first `<a>` element. Red box around the `<a></a>` sequence. Red arrow pointing from the `<a></a>` sequence to the XML tree diagram.

Exercise 1 - XML documents and trees.

- Draw the XML tree that corresponds to the XML document
- For each node, write down the pre- and post-identifiers.



Exercise 2 - XPath expressions.

Write the XPath expressions that correspond to the queries below, and evaluate the XPath expressions on the given XML document. Use the pre-identifiers to denote the output nodes of the queries.

- select all **a** nodes that have a **b**-parent *// a[parent::b] output 9, 10*
- select all **b** nodes that have no preceding **b**-elements *// b[not(preceding::b)] output 4*
- select all **b** nodes that are leaves *// b[not(descendant::*)] with output 4, 5*
- select all nodes with more than one **b**-sibling.

```
- //a[parent::b] with output 9, 10
- //b[not(preceding::b)] with output 4
- //b[not(descendant::*)] with output 4, 5
- //*[(count(preceding-sibling::b)+count(following-sibling::b))>1] with output 3
```

Exercise 3 - Node tests. Explain what the following XPath queries do and show the pre-identifier of nodes that are selected by them:

- `//a[2]`
- `//a/*[preceding-sibling::a and preceding-sibling::b]`
- `//*[count(a)=count(b)]` *有相同数量的子节点 a, b*
- `//*[count(preceding::a)>3]` *前面有3个a及以上的元素*
- `//a[not(ancestor::b)] | //b[not(ancestor::a)]`

- a-elements that are in the second position, with output 6, 10
- those children of a-parents that have a preceding a-sibling and a preceding b-sibling, with output 5
- element nodes which have the same numbers of a- and b-children, with output 3, 4, 5, 7, 9, 10
- elements that have more than 3 preceding a-elements, with output 10
- a-elements that do not have any b-ancestors together with b-elements that do not have any a-ancestors, with output 1, 2, 3, 6, 7

Big Data Management

XPatch

<?xml version="1.0" encoding="UTF-8">
<a><a><a/><a/>

- Draw the XML tree that corresponds to the XML document.
- For each node, write down the pre- and post-identifiers.

1 (c) 11

2 b' 5

3 b a | a a ✓

x 6 7

$$1/a[2]$$

$\text{deg-01-self}^* / a [2]$

先选所有 a , 再看是否是第二位置的 a
 ↓
 只看 a , 第 2 个 a

```

<?xml version="1.0" encoding="iso-8859-1"?>
<catalog>
  <dvd year="1954">
    <movie>The Seven Samuarai</movie>
    <director>Akira Kurosawa</director>
    <actor>Toshiro Mifune</actor>
    <feature>English Subtitles</feature>
    <price>29.99</price>
  </dvd>
  <dvd year="2005">
    <movie>Finding Neverland</movie>
    <musthave>Yes</musthave>
    <actor>Johnny Depp</actor>
    <actor>Kate Winslett</actor>
    <price>39.95</price>
  </dvd>
  <dvd year="1994">
    <movie>Pulp Fiction</movie>
    <director>Quentin Tarantino</director>
    <actor>Bruce Willis</actor>
    <scratch>Yes</scratch>
    <feature>Making Of</feature>
    <price>9.99</price>
  </dvd>
</catalog>

```

Exercise 4 - XML document and tree. Draw the corresponding XML tree.

Exercise 5 - Xpath queries. Write the following in XPath:

- a. Select all the price elements.
- b. Select all the price and movie elements.
- c. Select the second dvd.
- d. Select the self node of the second dvd.
- e. Select the parent of the second dvd.
- f. Select the ancestors of the second dvd.
- g. Select the children of the second dvd.
- h. Select the attribute children of the second dvd.
- i. Select the descendant elements of the second dvd.

- j.** Select the descendant nodes of the second dvd.
- k.** Select the preceding elements of the second dvd.
 - l.** Select the preceding nodes of the second dvd.
- m.** Select the preceding sibling elements of the second dvd.
- n.** Select the preceding sibling nodes of the second dvd.
- o.** Select the following elements of the second dvd.
- p.** Select the following nodes of the second dvd.
- q.** Select the following sibling elements of the second dvd.
- r.** Select the following sibling nodes of the second dvd.